

Lab 7: A Scaling Curve and Int8

Fit a toy power law, then quantize a weight matrix and measure error.

Work in a **Jupyter** notebook with **PyTorch** and `matplotlib`. Do notes **7.1–7.3** first. No Hugging Face weights this week.

```
import torch
import matplotlib.pyplot as plt
```

1. Fit a toy scaling law

Construct compute C and loss $L = aC^{-b} + c$ plus a little noise. You will recover a, b, c .

```
torch.manual_seed(0)
a_true, b_true, c_true = 2.0, 0.15, 1.4
C = torch.logspace(2, 6, 16)          # 1e2 .. 1e6
L = a_true * C.pow(-b_true) + c_true
L = L + 0.02 * torch.randn_like(L)
```

1. Scatter $\log C$ versus L . Guess whether c is near 1.4 before you fit.
2. Fit with Adam on unconstrained parameters, but keep $a > 0$, $b > 0$ via softplus:

```
p = torch.nn.Parameter(torch.tensor([1.0, 0.1, 1.0])) # raw a,b,c
opt = torch.optim.Adam([p], lr=0.05)
for _ in range(2000):
    a, b, c = torch.nn.functional.softplus(p[0]), torch.nn.functional.softplus(p[1]), p[2]
    pred = a * C.pow(-b) + c
    loss = ((pred - L) ** 2).mean()
    opt.zero_grad(); loss.backward(); opt.step()
```

Print fitted a, b, c versus truth. Overlay `pred` on the scatter.

3. In one sentence: if you only observed the four smallest C , would c still be identified?

2. Affine int8-ish quantization

```
torch.manual_seed(1)
x = torch.randn(256)
qmin, qmax = -128, 127
```

1. Compute `scale = (x.max() - x.min()) / (qmax - qmin)` and `zp = qmin - x.min() / scale`. Quantize with

```
q = torch.clamp(torch.round(x / scale + zp), qmin, qmax)
xhat = (q - zp) * scale
```

Print MSE `(x - xhat).pow(2).mean()`. It should be small relative to `x.var()`.

2. Repeat after replacing `x` with `x.clone()`; `x[0] = 50.0` (one outlier). MSE should jump. Why do activation outliers break naive INT8?
 3. Histogram `x` and `xhat` on one figure.
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3. Write-up

Four sentences: what C^{-b} says about diminishing returns, why Chinchilla would not only grow a 's "model" without growing tokens, what scale and zero-point are for, and one reason speculative decoding is not the same as distillation.